CONTACT 02 / 01

RAG SUPPORT

Two models power the RAG pipeline.

Embedding model

Converts passages and questions into vectors for search.

Generation model

Uses the retrieved passages to write an answer.

Model selection

Compare retrieval and answer quality, latency, usage and cost.

> Codex is our engineering tool. Its model choice is separate from these two runtime roles.

Code `workbench/teaching/provider_connection.py:19`

CONTACT 02 / 02

RAG SUPPORT

What does 1,536 dimensions mean?

A vector is a list

1,536 dimensions means 1,536 numbers for each passage or question.

One compatible space

Use the same embedding model and dimension setting for both.

Storage and quality

More dimensions use more vector storage. Measure the retrieval benefit.

> Dimensions measure vector length. Tokens measure text.

Code `workbench/teaching/contact2/config.py:20`

CONTACT 02 / 03

RAG SUPPORT

Why chunk, and how should we split?

Retrieve the relevant passage, keep useful context and avoid sending the whole document.

Fixed-size

Split by a character or token limit, often with overlap. Simple; can cut across a rule.

Recursive

Try paragraph, line and word boundaries before smaller splits, within a size limit.

Semantic

Use meaning or topic changes to choose boundaries. More processing; quality needs testing.

> Demo: size-based, word-aware chunks · 320 characters · 64-character target overlap.

Code `workbench/teaching/contact2/indexing/chunking.py:60`

CONTACT 02 / 04

RAG SUPPORT

Similarity, top-k and hybrid search.

Cosine + threshold

Higher cosine means more similar. The cutoff rejects weak vector matches.

Top-k + hybrid

Top-k limits results. Hybrid combines vector and keyword rankings.

Access filters

Tenant and role checks restrict which passages can be returned.

> Demo: top-k 3 · cosine cutoff 0.4, uncalibrated. A hybrid score is not cosine.

Code `workbench/teaching/contact2/querying/retrieval.py:14`

CONTACT 02 / 05

RAG SUPPORT

Retrieved does not mean supported.

Relevance

A similar passage may still omit the rule the question needs.

Context budget

Instructions, history, tools, passages and output all need room.

Evidence checks

Check the source, version and passage behind the answer.

> A citation matters only when its text supports the claim.

Code `workbench/teaching/contact2/querying/answer.py:9`

CONTACT 02 / 06

CODEX ENGINEERING

Choose the model and reasoning effort.

Model

Match capability, speed and cost to the engineering task.

Effort

Low / Medium / higher allocate different reasoning budgets, where supported.

Evidence

Judge the patch and checks, not how long the model thought.

> More reasoning does not grant more permission. Newer does not necessarily mean costlier.

CONTACT 02 / 07

CODEX ENGINEERING

A clear task before an implementation.

Normal mode

Can plan and implement. Say “plan only, then wait” if you want a pause.

Plan mode

Investigate, run checks and review the plan. Implementation edits stay on hold.

Start implementation

Exit Plan mode and send your prompt, or click “Implement this plan”.

> An implementation prompt alone does not exit Plan mode.

CONTACT 02 / 08

CODEX ENGINEERING

Repository instructions and task context.

AGENTS.md

Shared project rules: architecture, commands, boundaries and checks.

Directory scope

Root guidance applies broadly. Applicable nested files add local instructions.

Task prompt

The specific change and acceptance criteria for this assignment.

> Keep required project rules in AGENTS.md. Keep the current assignment in the prompt.

CONTACT 02 / 09

CODEX ENGINEERING

Context, compaction and memory.

Context · this request

The instructions, conversation, files and tool results supplied to the model.

Compaction · this thread

Condenses earlier work to keep going within the context limit. Details can be lost.

Memory · future chats

When enabled, Codex recalls useful context from eligible earlier chats.

> Keep required rules in AGENTS.md and exact results in a handover.

CONTACT 02 / 10

CODEX ENGINEERING

Reuse the beginning. Process what changes.

First request

`Project rules + tools`

Then: the first task.

Next request

The same beginning can reuse computation.
New input is processed normally.

Check the result

Inspect cached input tokens. A repeated question does not guarantee a hit.

> Prompt caching saves computation. It does not save an answer or add memory.

CONTACT 02 / 11

CODEX ENGINEERING

Choose a model for the job.

GPT-6 Luna

Focused, repeatable work.
Small edits, extraction and summaries.

GPT-6.1 Sol

Complex coding at lower cost than Astra.
A practical default for this change.

GPT-6 Astra

The hardest tasks across steps and tools.
Use when stronger judgment matters.

> Compare the same task and checks. Model age alone tells us little about cost.

CONTACT 02 / 12

CODEX ENGINEERING

Reasoning effort and Fast are different.

Effort · how much reasoning

Light / Low → Medium → High and above.
More depth can take more time and tokens.

Fast · service speed

Runs a supported model faster at higher usage.
It does not lower reasoning effort.

Choose deliberately

Start with the recommended effort.
Use Fast when the wait matters.

> Same model, separate controls: reasoning depth and service speed.

CONTACT 02 / 13

CODEX ENGINEERING

Older does not always mean cheaper.

Credits, not US dollars

MODEL	INPUT	CACHED INPUT	OUTPUT
GPT-6.1 SOL	50	2.5	250
GPT-5.6 SOL	100	10	500

> Current Codex credit rates · Standard speed · per 1M tokens · checked 23 Oct 2026

CONTACT 02 / 14

CODEX ENGINEERING

Check the session. Keep the decisions.

/status

Inspect the session and reported usage. Available fields depend on the client.

Pages · ChatGPT Space

Editable documents for plans and notes. Shared where access is granted.

Repository documents

Version-controlled plans and handovers. AGENTS.md holds project instructions.

> A saved Page or document affects a task when its relevant content is supplied.

Code Session commands · Pages

CONTACT 02 / 15

CODEX ENGINEERING

Choose the boundary and the reviewer.

Ask for approval

Work within the workspace.
Ask before crossing configured boundaries.

Approve for me

Keep the workspace boundary.
Route eligible requests to automatic review.

Full access

Broader file and internet access.
Not needed for our page-number change.

> Sandbox = where actions may run. Approvals = when and who reviews them.

CONTACT 02 / 16

CODEX ENGINEERING

A skill is a reusable procedure.

Review a change

Inspect a real diff, check risks and report actionable findings.

Check a browser journey

Verify visible behavior with isolated test data and evidence.

Ingest a policy

Call the existing pipeline, verify the result and report what happened.

> AGENTS.md gives repository guidance. A skill describes how to perform a kind of task.

CONTACT 02 / 17

CODEX ENGINEERING

Built-in tools, MCP and application tools.

Built-in tools

Codex reads files, edits code and runs allowed commands.

External MCP

A connected server exposes operations, such as documentation search.

Application tools

Relay calls a shipment function while answering a support question.

> Inspect the available operation and its scope before invoking it.

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CODEX ENGINEERING

Loops need feedback and a finish line.

Tool loop

Model requests a tool → tool result returns → model continues.

Engineering loop

Reproduce → patch → test → inspect feedback → improve.

Goal continuation

Pursue a thread objective within lifecycle, permission and budget limits.

> Stop at success, a budget limit, missing approval or an external blocker.

CONTACT 02 / 19

CODEX ENGINEERING

From an isolated change to a reviewable PR.

Worktree + tests

Keep live edits isolated. Reproduce the issue and check behavior and boundaries.

Browser verification

Exercise the visible journey using disposable data and inspect its result.

Diff + PR

Review scope, checks and remaining limits before proposing the change.

> The sequence is plan → reproduce → test → patch → browser check → diff review → PR.

CONTACT 02 / 20

CODEX ENGINEERING

The same task can run non-interactively.

Bounded input

Supply a concrete objective, repository context and acceptance criteria.

Controlled execution

Keep permissions, budgets and stopping conditions explicit.

Reviewable output

Capture the result, diff and actual verification evidence.

> The prepared CLI run is executed evidence. SDK/cloud examples are reference material.